THE THREE SPEEDS OF PHYSICAL AI & THE LATENCY STARVATION BOTTLENECK

How multi-step action chunking bridges the gap between neural inference latency and 1 kHz motor control

SYSTEM 2: SEMANTIC INTENT (0.5 - 2 Hz)

VLM Cognition / Expiring Leases

- Workload: Vision-Language grounding
- Latency budget: **500 – 2000 ms**
- Output: Goal affordance lease *L_intent*
- Platform: Linux MPU / Cloud

SYSTEM 1.5: ACTION CHUNKING (20 - 50 Hz)

Diffusion Policy / Trajectory Decoders

- Workload: Multi-step trajectory synthesis
- Latency budget: **20 – 50 ms (P99 ≤ 45 ms)**
- Output: Action chunks *A_{t:t+H} (H=16)*
- Platform: Embedded NPU / GPU

SYSTEM 1: SAFETY REFLEX (1000 Hz)

Control Barrier Functions / PWM Driver

- Workload: Mathematical safety enforcement
- Latency budget: **< 1.0 ms (Zero tail jitter)**
- Output: Gate driver motor PWM currents
- Platform: Deterministic Bare-Metal MCU

❌SINGLE-STEP POLICY: LATENCY STARVATION & CHATTER

Infer a_0 | STARVE (49ms) | Infer a_1 | STARVE (59ms)

a_0 (1ms) a_1

- **Physical Defect:** Motor executes 1 step (1 ms), then starves for 49 ms.
- **Zero-Order Hold Jumps:** Discrete position steps command infinite acceleration.
- **Consequence:** Violent mechanical vibration, gearbox fatigue, coil overheating.

✓ SYSTEM 1.5 ACTION CHUNKING: CONTINUOUS EXECUTION RESERVOIR

Infer Chunk | Action Reservoir: [a_0, ..., a_15] (H=16 / 320ms)

- **Temporal Amortization:** 50 ms inference generates 320 ms trajectory.
- **Jitter Immunity:** Motor queue never starves across P99 latency spikes.
- **1 kHz Splines:** MCU evaluates C^2 quintic polynomials every 1 ms tick.